

PROGRAM

6th International Workshop on Statistical Analysis on Multi-Outcome data

June 29-July 1st, 2026, Bordeaux, FR

DAY1
June 29, 2026

8:30-9:00 *Welcome*

9:00-9:15 Introduction (...)

9:15-10:15 One invited KEYNOTE Speech / 50' talk + 10'discussion

Title Navigating the Crossroads of Statistics, Generative AI, and Genomic Health
Xihong Lin, Biostatistics, Harvard T.H. Chan School of Public Health

10:15-10:45 *Coffee break*

10:45-12:25 [4 Parallel invited SESSIONS] 20' talk + 5'discussion

Session 1: Deep learning in survival analysis

Organizer: Lei Liu lei.liu@wustl.edu

- Jon Steingrimsson jon_steingrimsson@brown.edu
Title: Deep Learning with Time-to-event Outcomes
- Xingqiu Zhao xingqiu.zhao@polyu.edu.hk
Title: Robust Estimation and Inference for Accelerated Failure Time Models via Double Machine Learning
- Peter X Song pxsong@umich.edu
Title: Neural Network Machine Regression (NNMR): A Deep Learning Framework for Uncovering High-order Synergistic Effects
- Kevin He kevinhe@umich.edu
Title: Flexible Deep Learning Techniques for Survival Analysis with Data Integration

Session 2: Early or intercurrent events in (long-term outcome) clinical trials

Organizer: Catherine Legrand catherine.legrand@uclouvain.be

- Philippe Lambert p.lambert@uliege.be
Title: Joint Modeling of Longitudinal Health-Related Quality of Life Data in the Presence of Competing Dropout Risks
- Georgios Kazantzidis georgios.kazantzidis@roche.com
Title: A Joint Multi-State Model for Causal Mediation Analysis adjusting for treatment switching in Oncology
- Luc Boone luc.boone@eortc.org
Title: Estimation of inverse probability of censoring weights using shared parameter joint models for longitudinal and survival outcomes.
- Tomasz Burzykowski tomasz.burzykowski@iddi.com

Title: Early-outcome-based interim analyses in randomized clinical trials with long-term clinical endpoints

Session 3: Trustworthy AI and Statistical Innovation for Complex Health Data

Organizer: Ying Wei
And TBD

yw2148@cumc.columbia.edu

- Ying Wei yw2148@cumc.columbia.edu

Title: Title: Time-varying Quantile Regression with Multi-outcome Latent Groups

- Kaizheng Wang kaizheng.wang@columbia.ca

Title: Quantifying Fidelity in AI Persona Simulations

- Nan Lin nlin@wustl.edu

Title: Trajectory Clustering via Spatial-Graph and LLM-Semantic Embeddings

- Jinbao Chen jinboche@upenn.edu

Title: A Semiparametric Method for Addressing Under-Diagnosis Using Electronic Health Record Data

Session 4: Joint models with Bayesian inference

Organizer: Sangita Kulathinal (Finland)

sangita.kulathinal@helsinki.fi

- Etienne Sebag etienne.sebag@helsinki.fi

Title: Bayesian estimating Transition Rates in Two-State Non-Homogeneous Markov Jump Processes with Intermittent Observations: An Honest Time Data Augmentation Approach

- Marion keriou marion.keriou@inserm.fr

Title: A Bayesian partition model for multivariate functional data: discovering biological pathways by modelling the time course of gene expression level

- Trinh Dong Huu trinh.dong@ncl.ac.uk

Title: Double inverse probability weighting for modeling sparsely measured longitudinal markers in presence of left and right truncation by death

- Satrajit Roychoudhury satrajit.roychoudhury@pfizer.com

Title: Robust dynamic borrowing designs for randomized basket trials: a case study from the ultra-rare invasive mold infections

12:25-13:30 Lunch and POSTERS SESSION

13:30-15:10 [4 Parallel invited SESSIONS] 20' talk + 5'discussion

Session 5: Some Advanced Statistical Learning Methods for Modern Scientific Data

Organizer: Lexin Li

lexinli@berkeley.edu

And chair TBD

- Ji Zhu jizhu@umich.edu

Title: Modeling non-uniform hypergraphs using determinantal point processes.

- Yuhua Zhu yuhua.zhu@stat.ucla.edu

Title: Structure-aware model-free algorithms for continuous-time reinforcement learning.

- Moo Kyung Chung mkchung@wisc.edu

Title: Covariate-adjusted topological inference and learning.

- Junwei Lu junweilu@hsph.harvard.edu

Affiliation: Ranking inference for human feedback tuning in large language models.

Session 6: Joint models for risk prediction according to marker variability

Organizer: Helene Jacqmin-Gadda Helene.Jacqmin-Gadda@u-bordeaux.fr

- Jessica Barrett jessica.barrett@mrc-bsu.cam.ac.uk
Title: Flexible Bayesian semi-parametric approaches for modelling within-individual variability in joint models
- Leonie Courcoul leonie.courcoul@u-bordeaux.fr
Title: Joint models with heteroscedastic residual variance: an application to the impact of blood pressure variability on competing health events.
- Jianxin Pan, China jianxinpan@uic.edu.cn
Title: Measuring Biomarker Variability for Survival Data
- Michael Elliott mrelliot@umich.edu
Title: Joint Modeling of Multiple Longitudinal Biomarkers and Survival Outcomes via Threshold Regression: Variability as a Predictor

Session 7: Recent Developments in Survival Analysis

Organizer: Yifan Cui yifanc095@gmail.com

- Ingrid Van Keilegom ingrid.vankeilegom@kuleuven.be
Title: On an extension of the Cox model for time-dependent covariates under dependent censoring with unknown association
- Donglin Zeng dzeng@umich.edu
Title: Simultaneous Confidence Bands for Time-Varying Hazard Ratios with Interval-Censored Data
- Danyu Lin lin@bios.unc.edu
Title: Checking the Cox Proportional Hazards Model with Interval-Censored Data
- Zhe Fei zhfe@ucr.edu
Title: Adaptive Mixture-of-Experts Transformers for Joint Modeling of Irregular Longitudinal and Survival Data

Session 8: Trustworthy analysis of healthcare and medical data

Organizer: Fang Liu Fang.Liu.131@nd.edu
And Boris Hejblum boris.hejblum@u-bordeaux.fr

- Chuang Hong chuan.hong@duke.edu
Title: An Agentic AI System for Multi-Framework Communication Coding
- Yajuan Si yajuan@umich.edu
Title: Towards Differentially Private Finite Population Estimation: An Approach Based on Survey Weight Regularization
- Fang Liu Fang.Liu.131@nd.edu
Title: A Differentially Private Weighted Empirical Risk Minimization Procedure and its Application to Outcome Weighted Learning
- Xinghua Mindy Shi mindyshi@temple.edu
Title: Mathematical Foundations of Generative AI for Human Genetics and Epigenetics

Coffee break

15:40-17:20 [4 Parallel invited SESSIONS] 20' talk + 5'discussion

Session 9: Recent developments in reinforcement learning methods for precision medicine

Organizer: Donglin Zeng dzeng@umich.edu

- Alex Luedtke aluedtke@uw.edu

Title: DoubleGen: Debiased Generative Modeling of Counterfactuals

- TBD

- Nilanjana Laha nilanjanaaa.laha@gmail.com

Title: Model-free dynamic treatment regimes with arbitrary number of treatments and stages

- Yuanjia Wang yw2016@cumc.columbia.edu

Title: Learning Optimal Early Decision Treatment Rules with Multi-domain Intermediate Outcomes

Session 10: Distributional and Quantile Regressions for complex response data

Organizer: Antoine Barbieri antoine.barbieri@u-bordeaux.fr

- Angelo Alcaraz angelo.alcaraz@ens-paris-saclay.fr

Title: On asymmetric Laplace regression models: application to trophic diversity

- Lei Liu lei.liu@wustl.edu

Title: Efficient Estimation in Quantile Mixed Models via Smooth Check-Loss Approximation

- Mouna Abed mouna.abed@u-bordeaux.fr

Title: A joint model based on distributional regression to study the link between blood pressure and the risk of cerebral vasospasm.

- Gauthier Thurin Gauthier.thurin@ens.fr

Title: Optimal-transport based conformal prediction

Session 11: Innovative Statistical Methods for Complex Health and Biomedical Data

Organizer: Mei Hao hao.mei@ruc.edu.cn

- Hao Zhang haozhang@arizona.edu

Title: A Performance-Based Framework for Transfer Learning Measurement and Guidance

- Kai Zhang zhangk@email.unc.edu

Title: BELIEF in Dependence

- Lexin Li lexinli@berkeley.edu

Title: Brain Encoding and Decoding: Some Examples

- Hao Mei hao.mei@ruc.edu.cn

Title: Functional Latent Space Model for Time-Varying Networks: Application to Taiwan Administrative Claims Data

Session 12: High-dimensional surrogate markers in multi-outcome settings

Organizer: Boris Hejblum boris.hejblum@u-bordeaux.fr

- Layla Parast parast@austin.utexas.edu

Title: Resilience Measures for the Surrogate Paradox

- Florian Stijven florian.stijven@kuleuven.be

Title: Meta-Analytic Evaluation of Complex Surrogate Endpoints based on the Surrogate Index

- Tianxi Cai tcgai@hsph.harvard.edu

Title: Stable Multi-Surrogate Transformation for Robust and Generalizable Surrogacy

- Peter Gilbert peterg@uw.edu

DAY2 -
June 30, 2026

Welcome

[4 Parallel invited SESSIONS] 20' talk + 5'discussion

Organizer: Lan Luo

- Rui Duan rduan@hsph.harvard.edu

- Emily C Hector ehector@umich.edu

- Tian Gu tg2880@cumc.columbia.edu

- Ling Zhou zhouling@swufe.edu.cn

Session 14: Clustering from multivariate data

anais.rouanet@u-bordeaux.fr

And TBD

Title: Clustering of Multivariate Longitudinal Data of mixed type

- Julie Fendler julie.fendler@mrc-bsu.cam.ac.uk

Title: Consensus Monte Carlo for mixtures of categorical distributions

- Anais Rouanet anais.rouanet@u-bordeaux.fr

Title: Non-parametric clustering of multivariate longitudinal data with variable selection: Identifying sub-phenotypes of Alzheimer's Disease

- Qendresa Selimi qendresa.selimi@manchester.ac.uk

Title: A Dynamic Joint Latent Class Model With Time-Dependent Transition Probabilities

Session 15: The use of functional data analysis in multivariate settings

helene.jacqmin-gadda@u-bordeaux.fr

- Luo Xiao lxiao5@ncsu.edu

Title: Functional joint model for longitudinal and survival data with a functional covariate with application to a longitudinal study of objective physical activity

- Andrew Simpkin

andrew.simpkin@universityofgalway.ie

Title: Estimation and Application of Derivative Multivariate Functional Principal Component Analysis

- Robin Genuer robin.genuer@u-bordeaux.fr

Title: Random Survival Forest for the prediction of survival outcomes from time-varying outcomes using functional principal component analysis

- Sophie Dabo sophie.dabo@univ-lille.fr

Title: Preliminary Test Estimation in ULAN models for Functional Data

Session 16: New advances in survival analysis and machine learning

Organizer: Yichuan Zhao yichuan@gsu.edu
And TBD

- Edsel Pena pena@stat.sc.edu

Title: Prediction Machines Based on Concordance Correlation

- Yifan Cui yifanc095@gmail.com

Title : Double Machine Learning of Continuous Treatment Effects with General Instrumental Variables

- Liang Li lli15@mdanderson.org

Title: Modeling the Association Between Multivariate Nonlinear Longitudinal Data and Survival Outcome: A Measurement Error Perspective

- Yichuan Zhao yichuan@gsu.edu

Title: Novel empirical likelihood method for the cumulative hazard ratio under stratified Cox models

Coffee break

11:10-12:50

[4 Parallel invited SESSIONS] 20' talk + 5'discussion

Session 17: Statistical Learning for Complex Data

Organizer: Xingqiu Zhao xingqiu.zhao@polyu.edu.hk

- Jinfeng Xu jinfenxu@cityu.edu.hk

Title: Causal inference with outcome dependent sampling and mismeasured outcome

- Qixian Zhong qxzhong@xmu.edu.cn

Title: Semiparametric Inference for Functional Survival

- Qiang Wu qiangwu@mail.bnu.edu.cn

Title: Deep Nonparametric Inference for Interval-Censored Data

- Xiangbin Hu xiang-bin.hu@connect.polyu.hk

Title: Deep Conditional Generative Learning for Optimal Individualized Treatment Rules

Session 18: Software for the Analysis of Multi-Outcome Models (INLAjoint, JMBayes2, frailtypack ...)

Organizer: Denis Rustand denis.rustand@u-bordeaux.fr

- Elias Krainski elias.krainski@kaust.edu.sa

Title: graphpcor: Models for correlation matrices based on graphs.

- Virginie Rondeau Virginie.Rondeau@inserm.fr

Title: Assessing surrogacy from joint modeling and mediation analysis when surrogates are either censored event times or longitudinal biomarker: cancer application

- Janet van Niekerk Janet.vanNiekerk@up.ac.za
Title: INLAjoint and beyond: INLA for multi-outcome modeling.

- Pedro Afonso p.mirandaafonso@erasmusmc.nl
Title: The JMBayes2 R package for joint models of longitudinal and time-to-event data: From flexible fitting to dynamic prediction and validation

Session 19: Advanced Analytical Methods in Multi-outcome Data
Organizer: Lei Liu lei.liu@wustl.edu

- Yun Li Yun.Li@Pennmedicine.upenn.edu
Title: Confounder Missingness in EHR Comparative Effectiveness: MI or PS Calibration?
- Grace Y. Yi gyi5@uwo.ca
Title: Function-on-Scalar Regression with Ultra high-Dimensional Error-Prone Covariates
- Douglas Schaubel douglas.schaubel@pennmedicine.upenn.edu
Title: Time-varying frailty model for analyzing recurrent/terminal event data
- Yi Li yili@umich.edu
Title: Inference for Deep Neural Network Estimators

Session 20: Causal Learning for Optimal Health Decision-Making with Complex Outcomes
Organizer: Yuanjia Wang yw2016@cumc.columbia.edu

- Erica Moodie erica.moodie@mcgill.ca
Title: Estimating the optimal allocation of kidneys in the presence of competing risks from clustered data
- Ashken Ertefie Ashkan.Ertefaie@PennMedicine.upenn.edu
Title: A structural nested rate model for estimating the effects of time-varying exposure on recurrent event outcomes
- Yuan Chen chenyl9@mskcc.org
Title: Learning Locally Interpretable Individualized Treatment Rules
- Cheng Zheng cheng.zheng@unmc.edu
Title: Joint Model for Mediation Analysis with Causally Related Longitudinal and Recurrent Event Mediators for Survival Outcome.

13:00-14:00

Lunch and POSTERS SESSION

14:00-15:40

[4 Parallel invited SESSIONS] 20' talk + 5'discussion

Session 21: Statistical and machine learning methods for complex biomedical data

Organizer: QI Long, qlong@pennmedicine.upenn.edu
and Kevin He, kevinhe@umich.edu

- Hongzhe Li hongzhe@upenn.edu
Title: A Robust Local Frechet Regression Using Unbalanced Neural Optimal Transport with Applications to Dynamic Single-cell Genomics Data
- Sandra Safo ssafo@umn.edu
Title: iDeepViewLearn+: Interpretable AI for Robust and Fair Multimodal Data Integration
- Ali Shojaie ashojaie@uw.edu

Title: GLM Inference with AI-Generated Synthetic Data Using Misspecified Linear Regression

- Nicholas Henderson nchender@umich.edu

Title: Ranking-based Data Integration for Robust Outcomes Prediction and Biomarker Detection

Session 22: Causal inference in presence of truncation by death

Organizer: Cecile Proust-Lima cecile.proust-lima@u-bordeaux.fr

- Daniel Nevo danielnevo@gmail.com

Title: Causal effects on non-terminal event time with application to antibiotic usage and future resistance

- Catherine Legrand catherine.legrand@uclouvain.be

Title: Bias-Corrected Two-Stage Modeling for Joint Analysis of Multidimensional HRQoL and Survival

- Arce Domingo-Relloso ad3531@cumc.columbia.edu

Title: A path-specific effect approach to mediation analysis with time-varying mediators and time-to-event outcomes accounting for competing risks

- Stijn Vansteelandt. stijn.vansteelandt@ugent.be

Title: *Evaluating Treatment Effects in Longitudinal Trials with Truncation by Death: An Assumption-Lean Approach*

Session 23: Innovative Statistical Methods for Longitudinal and Survival Analysis in Medical Studies

Organizer: Yanqing Sun yasun@charlotte.edu

- Lang Wu lang@stat.ubc.ca

Title: Joint models in longitudinal studies for efficient and robust inferences

- Jimin Ding jmding@wustl.edu

Title: Enhanced survival trees with applications to Alzheimer's disease study

- Wenqing He whe@stats.uwo.ca

Title: Boosting methods for interval censored data with regression and classification

- Yanqing Sun yasun@charlotte.edu

Title: Estimation and Inference of Semiparametric Temporal Intensity Models for Recurrent Events with Application to a Malaria Vaccine Trial

Session 24: New Estimation and Inference Methods for Analyzing Composite Survival Outcomes

Organizer: TBD

- Chen Hu huc@jhu.edu

Title: On statistical inference of multiple competing risks in oncology clinical trials

- Yu Cheng yucheng@pitt.edu

Title: Win odds for Sequential Multiple Assignment Randomized Trials

- TBD

- Yingye Zheng yzheng@fredhutch.org

Title: Joint Modeling of True and Misclassified Event Times for Biomarker Evaluation

15:40-16:10

Coffee break

16:10-17:50	[4 Parallel invited SESSIONS] <i>20' talk + 5'discussion</i>
	Session 25: Machine Learning and Survival Analysis Organizer: Jiaoguo Sun sunj@missouri.edu • Yu Gu yugu@hku.hk Title: Semiparametric Functional Multi-State Models: Estimation and Testing with Application to Alzheimer's Disease • Jiaqi Zhao. jiaqi2025.zhao@connect.polyu.hk Title: Spline Learnable Neural Networks with Applications to Generative Diffusion Models • Li Liu liu.math@whu.edu.cn Title: Automatic structure identification and variable selection for additive accelerated failure time model with ultra high dimensional covariates • Rujia Cao rujia.cao@connect.polyu.hk Title: Deep Semiparametric and Nonparametric Inference for Longitudinal Data
	Session 26: Area of causal inference Organizer: Cheng Zheng cheng.zheng@unmc.edu • Ying Zhang ying.zhang@unmc.edu Title: Joint Analysis of Multivariate Longitudinal Data and Interval-Censored Event Time Data with Application to Huntington's Disease Progression • Ran Dai ran.dai@unmc.edu Title: Model-free High Dimensional Mediator Selection with False Discovery Rate Control • Mladen Kolar mkolar@marshall.usc.edu Title: Confidence Sets for Causal Orderings • Youngjoo Cho yvc5154@konkuk.ac.kr Title: Doubly robust inference for the heterogeneous treatment effect of multiple treatments on time-to-event outcomes
	Session 27: Joint analysis of time-to-event or longitudinal markers in the presence of mortality as a competing risk Organizer: Sebastien Haneuse shaneuse@hsph.harvard.edu • Eleni-Rosalina Andrinopoulou eleni.r.andrinopoulou@gsk.com Title: Challenges and Opportunities in Joint Modelling of Multiple Longitudinal and Survival Outcomes • Daniels, Michael J mdaniels@stat.ufl.edu Title: A Bayesian Nonparametric Approach for Semi-Competing Risks with Application to Cardiovascular Health • Sebastien Haneuse shaneuse@hsph.harvard.edu Title: A novel framework for Joint prediction of longitudinal markers and death • Sangita Kulathinal sangita.kulathinal@helsinki.fi Title: Nonparametric estimation of the joint and conditional survival functions of the time to an event of interest and associated integrated covariate processes
	Session 28 : Nonparametric and semiparametric modeling for the analysis of complex data Organizer: Zhezhen Jin zj7@cumc.columbia.edu

	- Shanshan Ding sding@udel.edu Title: Robust causal effect estimation in high-dimensional survival analysis using sufficient dimension reduction - Xuwen Lu xlu@ucalgary.ca Title: Variable Selection in the Joint Frailty Model of Recurrent and Terminal Events Using Broken Adaptive Ridge Regression. - Leilei Zeng lzensg@uwaterloo.ca Title: Modeling Heterogeneous Disease Processes Using a Mixture of Multistate Hidden Markov Models - Hua Shen hua.shen@ucalgary.ca Title: Integrative Analysis of Multiple Data Sources Involving Misclassification and Missing Data
19:45-23:00	Dinner / “Le Tchanqué” / Buffet / N=170 ?

DAY 3 - July 1st, 2026	
8:30-9:00	Welcome
9:00-10:40	[4 Parallel invited SESSIONS] 20' talk + 5'discussion
	Session 29: Machine or deep learning for multiple longitudinal and survival data Organizer: Virginie Rondeau Virginie.rondeau@inserm.fr - Manel Rakez rakez.manel@outlook.fr Title: Joint models and deep learning neural networks for the longitudinal analysis of mammography images in screen-detected breast cancer risk prediction - Agathe Guilloux agathe.guilloux@inria.fr Title: Deep Learning for Longitudinal and Survival Data: Toward Realistic Synthetic Patient Generation - Clemens Schaechter clemens.schaechter@uniklinik-freiburg.de Title: Modeling disease progression and treatment switches in rare disease trials by integrating statistical modeling and synthetic experts - Cécile Proust-Lima cecile.proust-lima@u-bordeaux.fr Title: Neural differential equations enhanced linear mixed model
	Session 30: Causal mediation pathway analysis Organizer: Peter Song pxsong@umich.edu - Jiasheng SHI shijiasheng@cuhk.edu.cn Title: Spillovers Through Time: A Neural Framework for Temporal Network Data and Dynamic Causal Effect Estimation - Yen-Tsung Huang ythuang@stat.sinica.edu.tw Title: Semiparametric mediation analysis using single-index models

- Zhonghua Liu zl2509@cumc.columbia.edu

Title: Mitigating Unmeasured Confounding Bias in Large-Scale Causal Mediation Analysis Via Factor Analysis in Epigenome-Wide Studies

- Wei Zhou zhouwei23@swufe.edu.cn

Title: Statistical Inference for Mediation Models with High Dimensional Exposures and Mediators

Session 31: Advanced uses of joint modeling techniques with application in longitudinal and time-to-event data

Organizer: qitris Rizopoulos. d.rizopoulos@erasmusmc.nl

- Jeremy Taylor jmgt@umich.edu

Title: Dynamic Predictions and Predictimands for Salvage Therapy in Recurrent Prostate Cancer Using Joint Models

- Esra Kurum esra.kurum@ucr.edu

Title: Scalable Time-Dynamic Joint Models for Multilevel Health Data

- Christos Thomadakis cthomadak@aueb.gr

Title: Shared parameter modeling of longitudinal data allowing for possibly informative visiting process and terminal event

- Eleanor Pullenayegum eleanor.pullenayegum@sickkids.ca

Title: Addressing measurement times that are not at random in longitudinal data through joint modelling of outcomes and latent disease processes

Session 32: Tensor-Train Methods, Partial Derivatives, and High-dimensional Inference in Modern Statistics

Organizer: Yong Chen. ychen123@pennmedicine.upenn.edu

- Yong Chen ychen123@pennmedicine.upenn.edu

Title: PDA: a unified framework for lossless, one-shot, federated learning algorithms

- Jingmei Qiu jingqiu@udel.edu

Title: Projection-Enhanced Interpolation-Based Tensor-Train Decompositions for One-Shot, Privacy-Preserving Distributed Inference

- Xiaowu Dai dai@stat.ucla.edu

Title: Statistical Learning via Partial Derivatives

- Yumou Qiu. qiuyumou@math.pku.edu.cn

Title: High-dimensional Data Assimilation for Large Scale Physical Systems

10:40-11:10

Coffee break

11:10-12:10

One invited KEYNOTE Speech / 50' talk + 10'discussion

Title Harnessing Variational Auto-Encoders for Fitting Deep Mixed-Effects Models

Dimitris Rizopoulos, Erasmus University Medical Center
d.rizopoulos@erasmusmc.nl

12:10-12:30

Closing remarks